

Qatar-1b

R-2727

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_130908 · created 2026-09-01T23:45:39+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456543.66861 +/- 0.00011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.057
Transit depth (Rp/Rs)^2	3.65 +/- 2.18 [%]
Orbital Inclination (inc)	88.87 +/- 2.88
Airmass coefficient 1 (a1)	0.03 +/- 0.017
Airmass coefficient 2 (a2)	2.94 +/- 1.71
Scatter in the residuals of the lightcurve fit is	61.03 %
Best Comparison Star	#2 - [551, 426]
Optimal Aperture	2.29
Optimal Annulus	7.25
Transit Duration (day)	0.0833 +/- 0.0071

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.191 $\pm$ 0.057 vs 0.1463 (deviation 0.78 σ)
Duration fit vs published	0.0833 d vs 0.0692 d (deviation 1.99 σ)
Mid-time O-C	0.2 min
Depth SNR	3.4
Residual scatter	61.03 %

Light curve

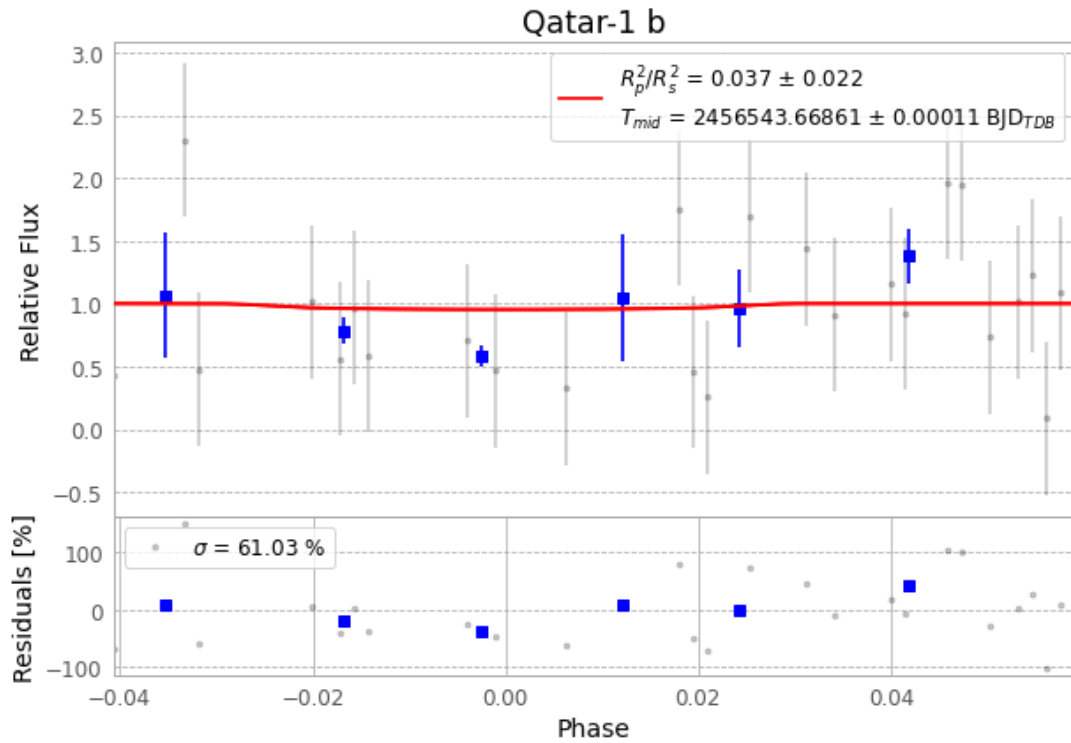


Figure 1 — FinalLightCurve_Qatar-1 b_2013-09-07.png (SHA-256 0ff1113318b1e409...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [337, 184], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6ebcfa0a58fcba89...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), Qatar-1130908023615.FITS → Qatar-1130908060012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-130908.log (30811abbc041...), FinalLightCurve_Qatar-1 b_2013-09-07.png (0ff1113318b1...), FinalLightCurve_Qatar-1 b_2013-09-07.csv (ca2fb8ff9e5f...)

Compute resources

Wall time 391.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 23:45Z.